

Mohamed Saber

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1 Research Interests

Hardware-Software Co-Design, Efficient Deep Learning, Model Optimization, Neuromorphic & Edge Architectures.

2 Education

M.Sc. in Embedded Artificial Intelligence

Ibn Zohr University

Agadir, Morocco

2024 – 2026

- Exchange Semester, Electrical Engineering | University of Twente (Grade: 9.5/10 – Top 7% of cohort)
- **Focus:** TinyML, Fog and Edge Computing, On-Chip Memory Optimization
- **Thesis Topic:** Scaling a Vision Transformer Block on SparkRV Many-Core Neuromorphic Processor
- **Supervisor:** Dr. ir. Amirreza Yousefzadeh (UTwente) - a.yousefzadeh@utwente.nl

B.Sc. in Computer Engineering

Ibn Zohr University

Agadir, Morocco

09/2020 – 07/2023

- **Focus:** Computer Systems, Cloud Computing, Software Architecture, and Data Structures

3 Professional Experience

Research Intern

CAES Research Group, University of Twente

Enschede, Netherlands

02/2026 – 08/2026

- Mapped DeiT Vision Transformer blocks onto SparkRV (open-source RISC-V neuromorphic multi-core processor) under a strict 256 KB per-core SRAM budget.
- Implemented custom firmware operators (in-place Softmax, GELU), reducing memory footprint by 29%, and engineered multi-core NoC distribution across 31- and 43-core topologies.
- Evaluated full Performance-Power-Area (PPA) trade-offs, achieving $14.73\times$ throughput improvement via asymmetric compute scaling at 22nm.

AI Software Engineer (Contract)

FandaSoft

Marrakech, Morocco

07/2025 – 10/2025

- Automated custom DSL generation using LLMs through AWS Bedrock, and eliminated manual development.
- Engineered iteration loops and testing frameworks to validate model outputs in production environments.
- Optimized few-shot prompt engineering workflows and data preprocessing pipelines to improve model outputs.

Founding Software Engineer

Yutapp

Arlington, USA (Remote)

02/2023 – 02/2025

- Co-designed a scalable, multi-tenant SaaS backend architecture and led the development of a core website builder architecture.
- Built modular web services and data pipelines, collaborating across an international remote engineering team.

4 Projects

AskUiz – University RAG Chatbot

[\[Demo Link\]](#)

- Built a Retrieval-Augmented Generation (RAG) system using vector databases and semantic search over university documentation, to answer students' inquiries.

5 Programming Skills

- **Languages (proficient):** Python, C/C++, TypeScript
- **Languages (familiar):** SQL, Java
- **Machine Learning & Frameworks:** PyTorch, TensorFlow, Hugging Face, Scikit-learn
- **Systems & Tools:** Linux, Git, Docker, AWS, RISC-V Toolchain
- **Spoken Languages:** English (IELTS Academic – B2), French (Fluent), Arabic (Native)